

T8

$n=200$ $H_0: \xi$ и η незав. $H_1: H_0 \quad \alpha=0,05$.

$\eta:$	$\xi:$ зачисл. 25	норм. зачисл. 50	зачисл. 25	$q_i(\eta)$ $\frac{100}{200}$
1				
2	52	41	7	$\frac{700}{200}$
$P_i(\xi)$	$\frac{27}{200}$	$\frac{91}{200}$	$\frac{39}{200}$	

$$\begin{aligned} \tilde{\Delta} = & \frac{(125 - 200 \cdot \frac{77}{200} - \frac{1}{2})^2}{200 \cdot \frac{77}{200} \cdot \frac{1}{2}} + \frac{(500 - 200 \cdot \frac{91}{200} - \frac{1}{2})^2}{200 \cdot \frac{91}{200} \cdot \frac{1}{2}} + \frac{(125 - 200 \cdot \frac{33}{200} - \frac{1}{2})^2}{200 \cdot \frac{33}{200} \cdot \frac{1}{2}} + \\ & + \frac{(52 - 200 \cdot \frac{77}{200} - \frac{1}{2})^2}{200 \cdot \frac{77}{200} \cdot \frac{1}{2}} + \frac{(41 - 200 \cdot \frac{91}{200} - \frac{1}{2})^2}{200 \cdot \frac{91}{200} \cdot \frac{1}{2}} + \frac{(7 - 200 \cdot \frac{33}{200} - \frac{1}{2})^2}{200 \cdot \frac{33}{200} \cdot \frac{1}{2}} \approx 20,206. \end{aligned}$$

$$\Delta \rightsquigarrow \chi^2(2)$$

$$p\text{-value} = P(\Delta \geq \tilde{\Delta} | H_0) = \int_{20,206}^{\infty} q(t) dt \approx 4,1 \cdot 10^{-5} < \alpha \rightarrow \text{omb. } H_0$$